



Module 6: Data Link Layer

Telecom Academy
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Introduction to Networks v7.0
(ITN)



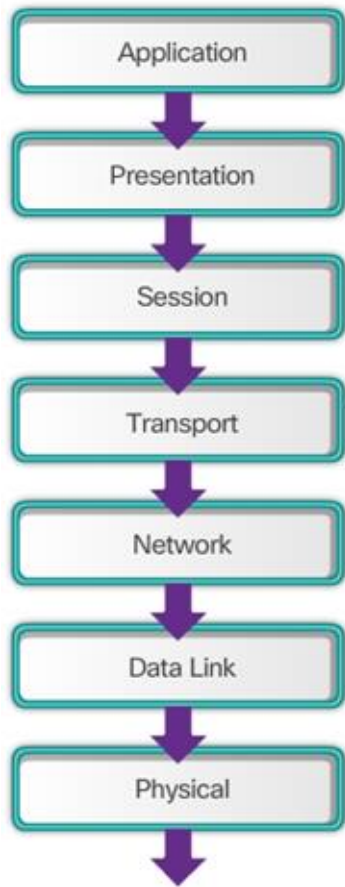
Module Objectives

Module Title: Data Link Layer

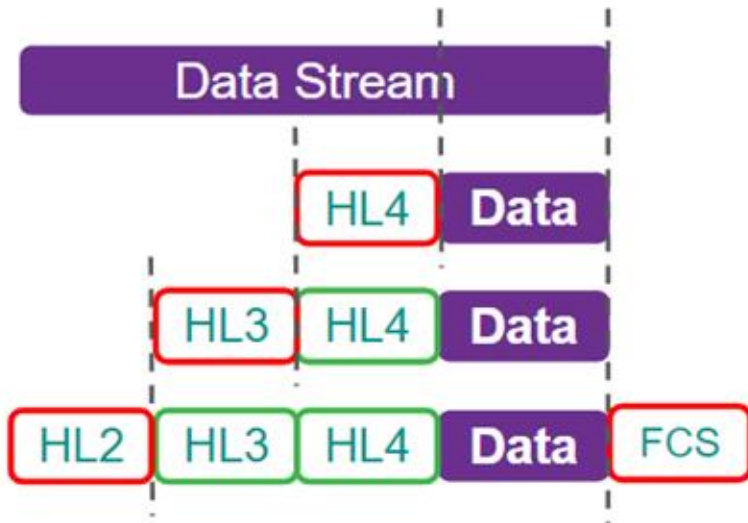
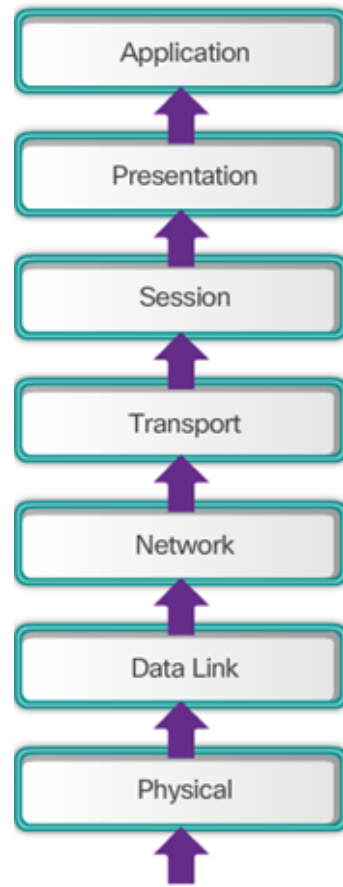
Module Objective: Explain how media access control in the data link layer supports communication across networks.

Topic Title	Topic Objective
Purpose of the Data Link Layer	Describe the purpose and function of the data link layer in preparing communication for transmission on specific media.
Topologies	Compare the characteristics of media access control methods on WAN and LAN topologies.
Data Link Frame	Describe the characteristics and functions of the data link frame.

Source Node



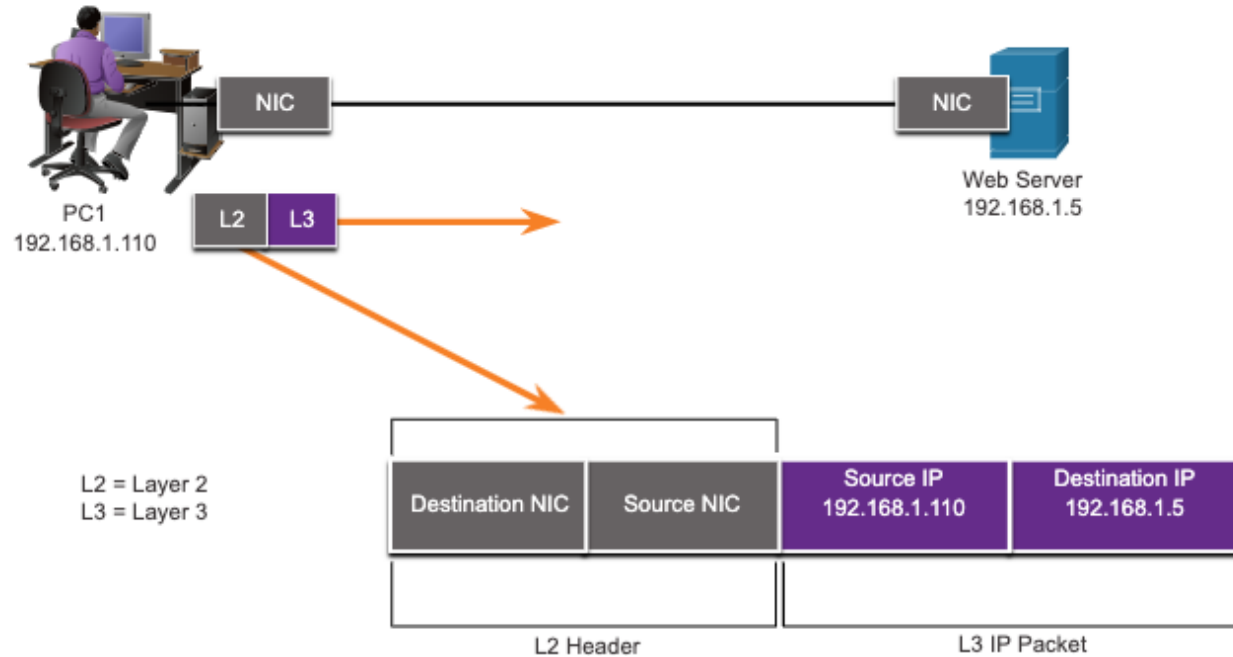
Destination Node

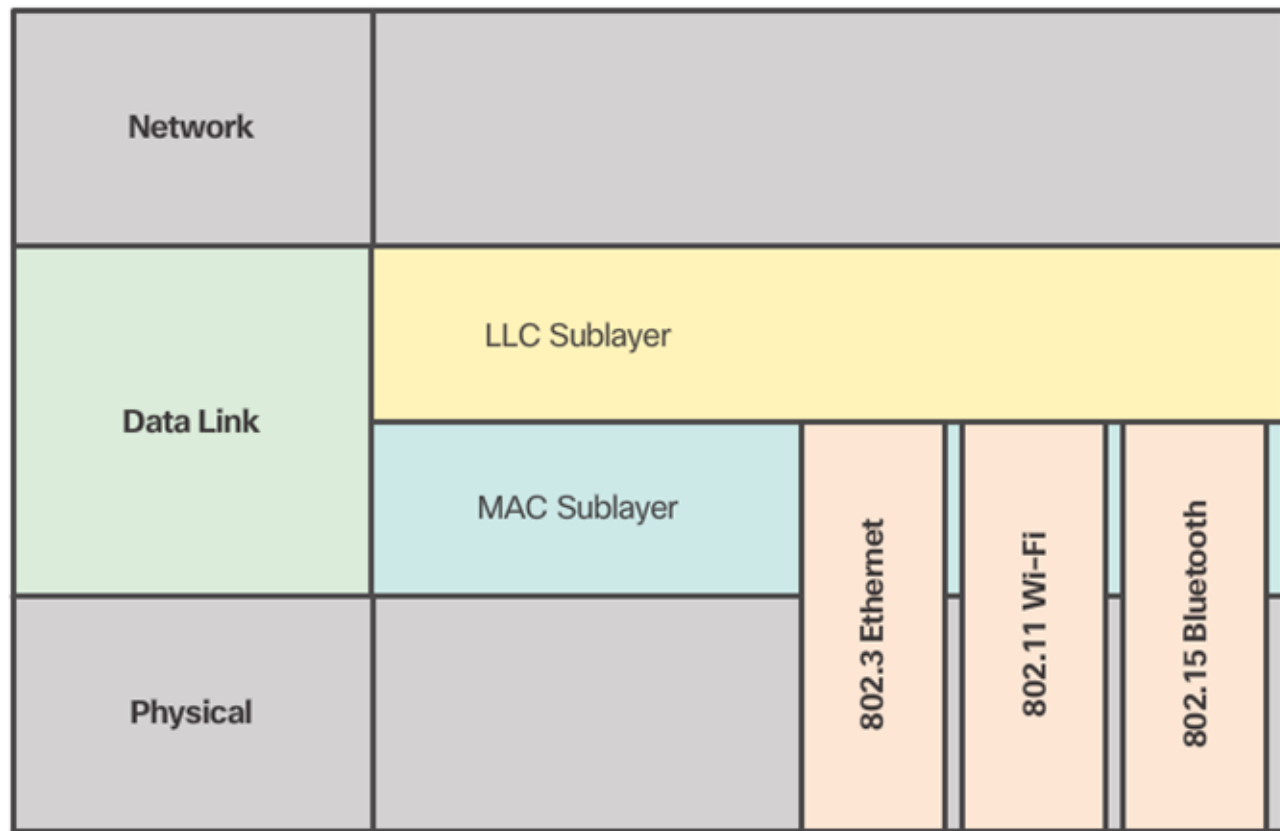


6.1 Purpose of the Data Link Layer

Purpose of the Data Link Layer

The Data Link Layer

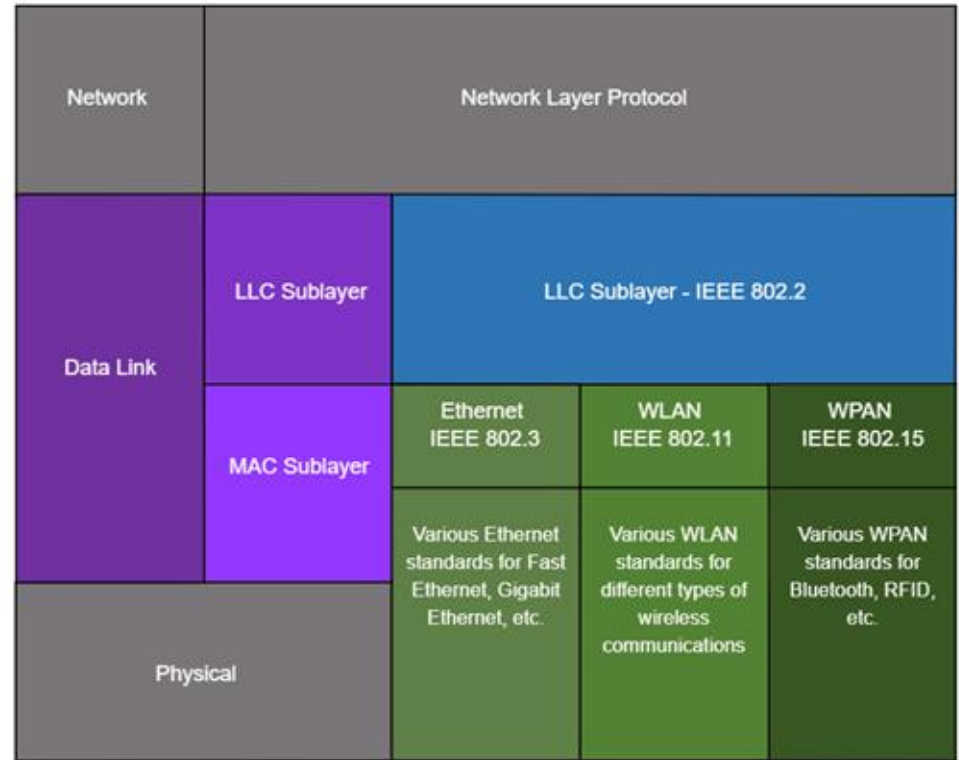




IEEE 802 LAN/MAN Data Link Sublayers

The Data Link Layer consists of two sublayers. **Logical Link Control (LLC)** and **Media Access Control (MAC)**.

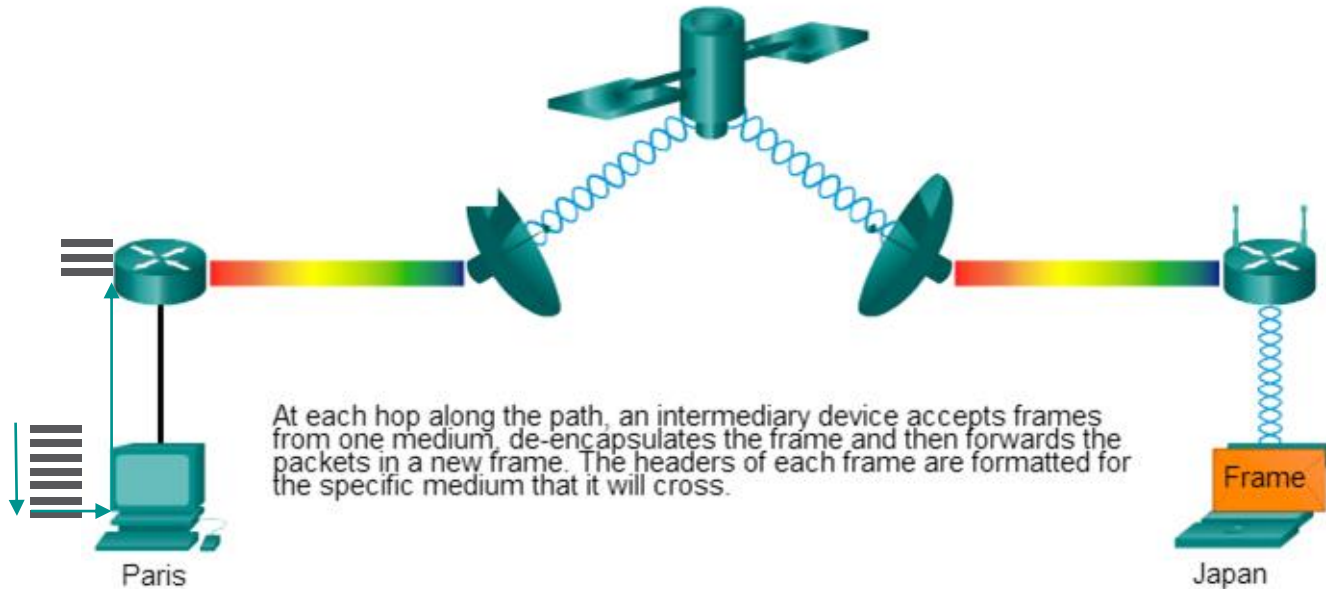
- The LLC sublayer communicates
- The MAC sublayer is responsible for data encapsulation and media access control.

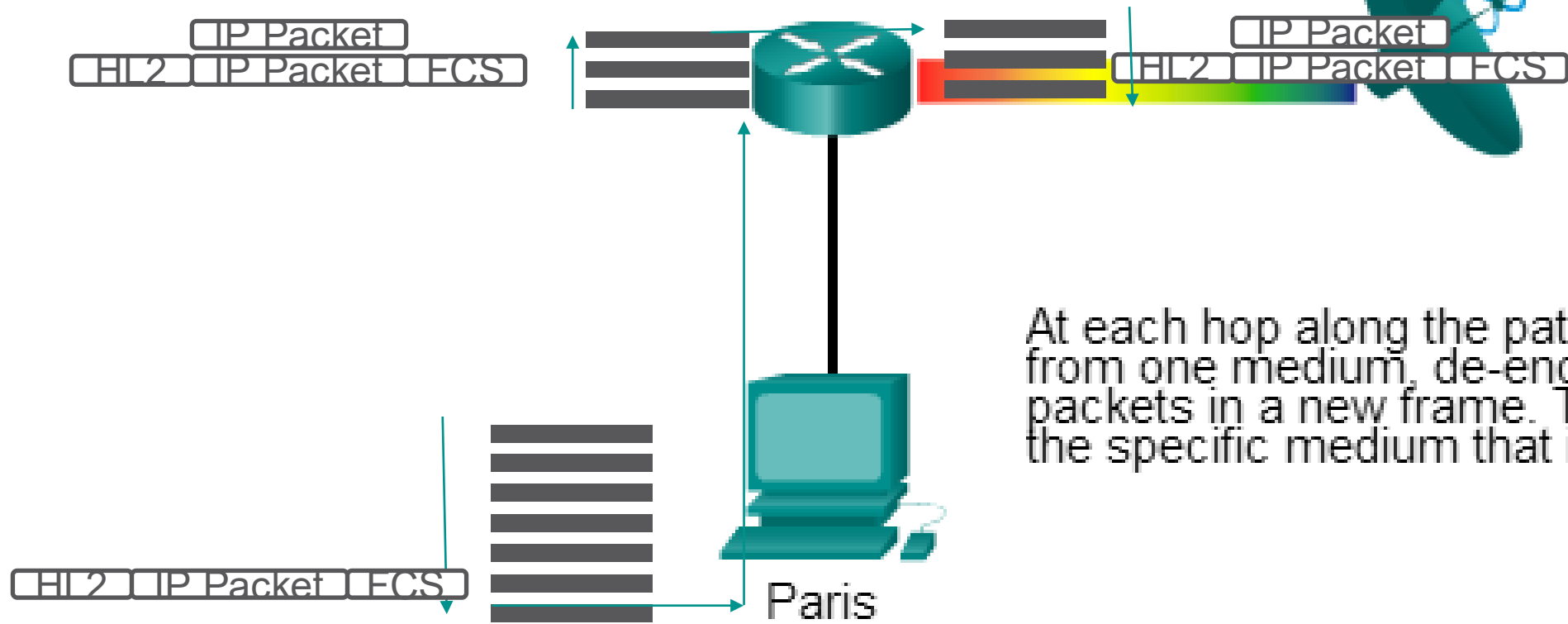


Media Access Control

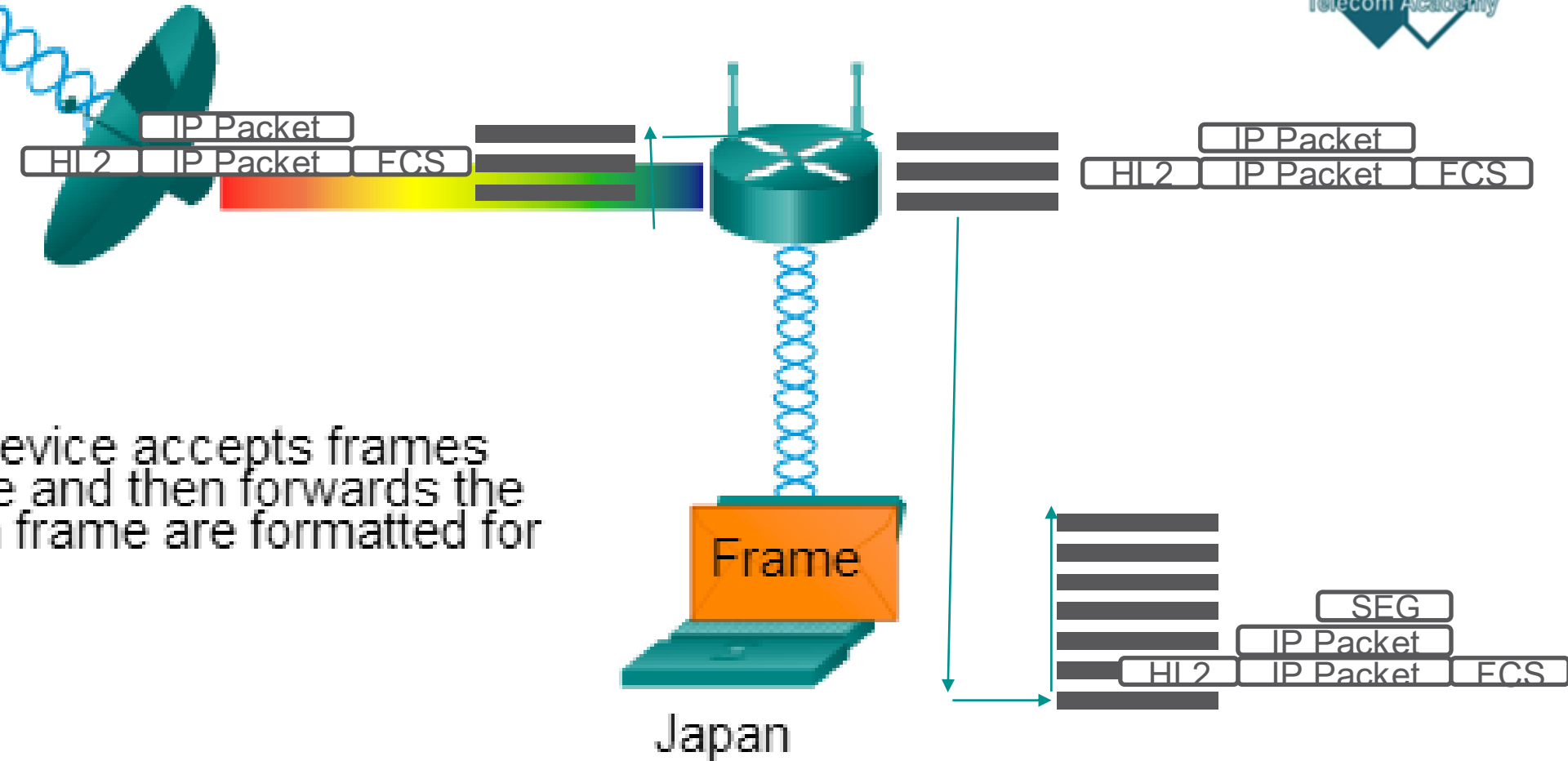
Data link layer protocols govern how to format a frame for use on different media.

Different protocols may be in use for different media.





At each hop along the path, the router de-encapsulates the frame from one medium, de-encapsulates the packets in a new frame. The router then encapsulates the packets in the specific medium that it is sending them to.



Device accepts frames
and then forwards the
frame are formatted for

6.2 Topologies



Topologies

Physical and Logical Topologies

There are two types of topologies used when describing networks:

- **Physical topology** – shows physical connections and how devices are interconnected.
- **Logical topology** – identifies the virtual connections between devices using device interfaces and IP addressing schemes.

Topologies

WAN Topologies

There are three common physical WAN topologies:

- **Point-to-point**
- **Hub and spoke**
- **Mesh**

Topologies

Point-to-Point WAN Topology

Physical point-to-point topologies directly connect two nodes.

The nodes may not share the media with other hosts.

Because all frames on the media can only travel to or from the two nodes, Point-to-Point WAN protocols can be very simple.



Topologies

LAN Topologies

End devices on LANs are typically interconnected using a star or extended star topology. Star and extended star topologies are easy to install, very scalable and easy to troubleshoot.

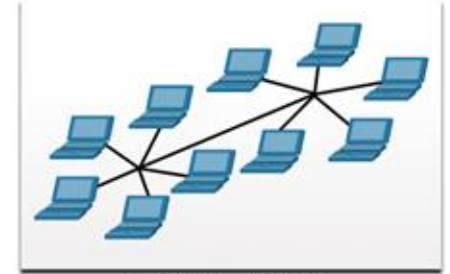
Early Ethernet and Legacy Token Ring technologies provide two additional topologies:

- **Bus** – All end systems chained together and terminated on each end.
- **Ring** – Each end system is connected to its respective neighbors to form a ring.

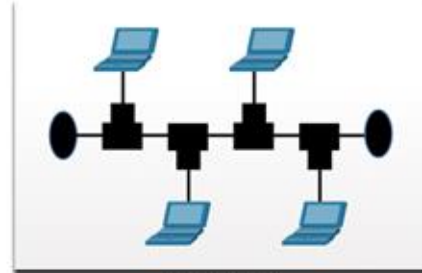
Physical Topologies



Star Topology



Extended Star Topology



Bus Topology



Ring Topology

Topologies

Half and Full Duplex Communication

Half-duplex communication

- Only allows one device to send or receive at a time on a shared medium.
- Used on WLANs and legacy bus topologies with Ethernet hubs.

Full-duplex communication

- Allows both devices to simultaneously transmit and receive on a shared medium.
- Ethernet switches operate in full-duplex mode.

Topologies

Access Control Methods

Contention-based access

All nodes operating in half-duplex, competing for use of the medium. Examples are:

- Carrier sense multiple access with collision detection (CSMA/CD) as used on legacy bus-topology Ethernet.
- Carrier sense multiple access with collision avoidance (CSMA/CA) as used on Wireless LANs.

Controlled access

- Deterministic access where each node has its own time on the medium.
- Used on legacy networks such as Token Ring and ARCNET.

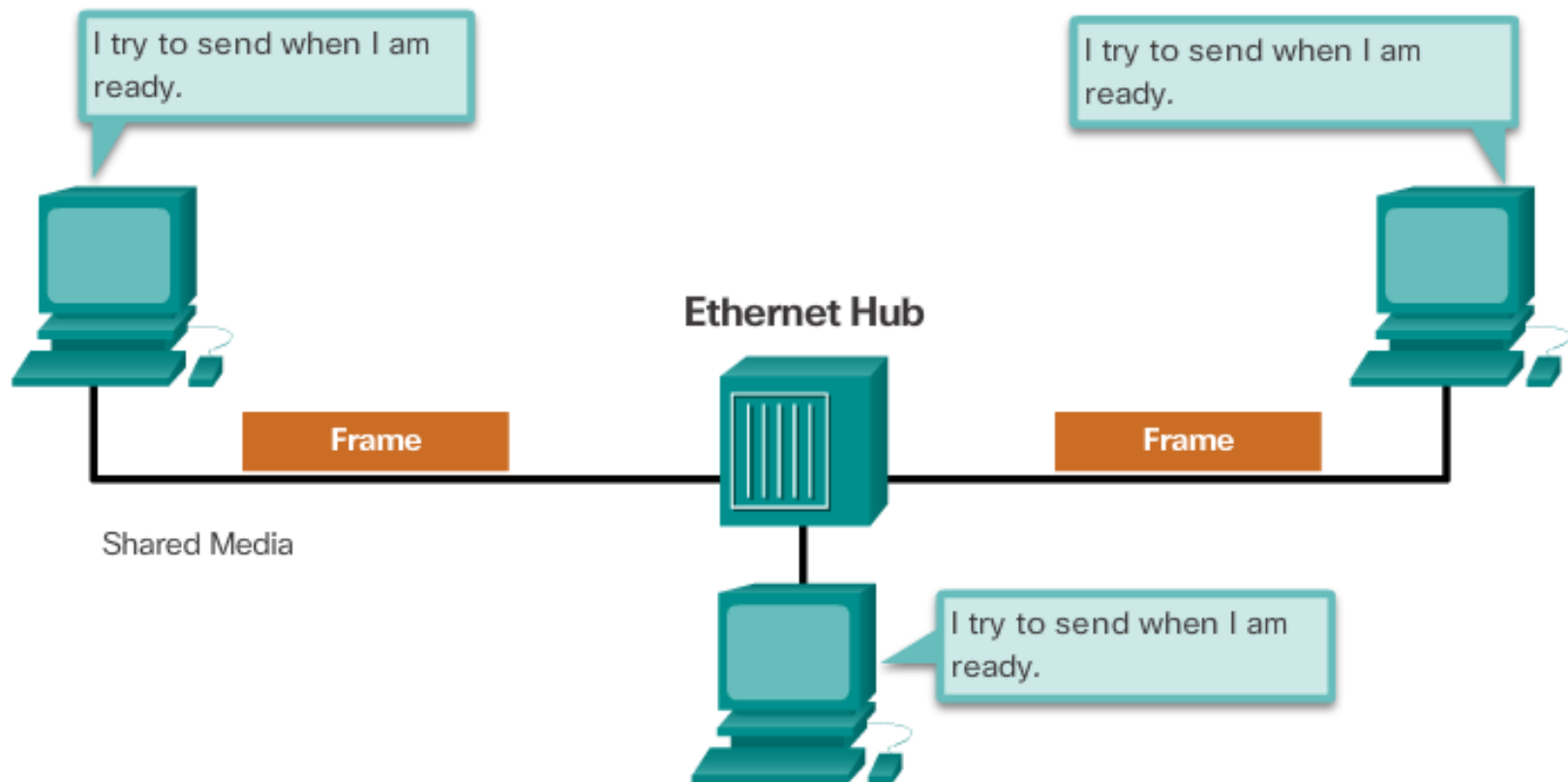
Contention-Based Access – CSMA/CD

CSMA/CD

- Used by legacy Ethernet LANs.
- Operates in half-duplex mode where only one device sends or receives at a time.
- Uses a collision detection process to govern when a device can send and what happens if multiple devices send at the same time.

CSMA/CD collision detection process:

- Devices transmitting simultaneously will result in a signal collision on the shared media.
- Devices detect the collision.
- Devices wait a random period of time and retransmit data.



Contention-Based Access – CSMA/CA

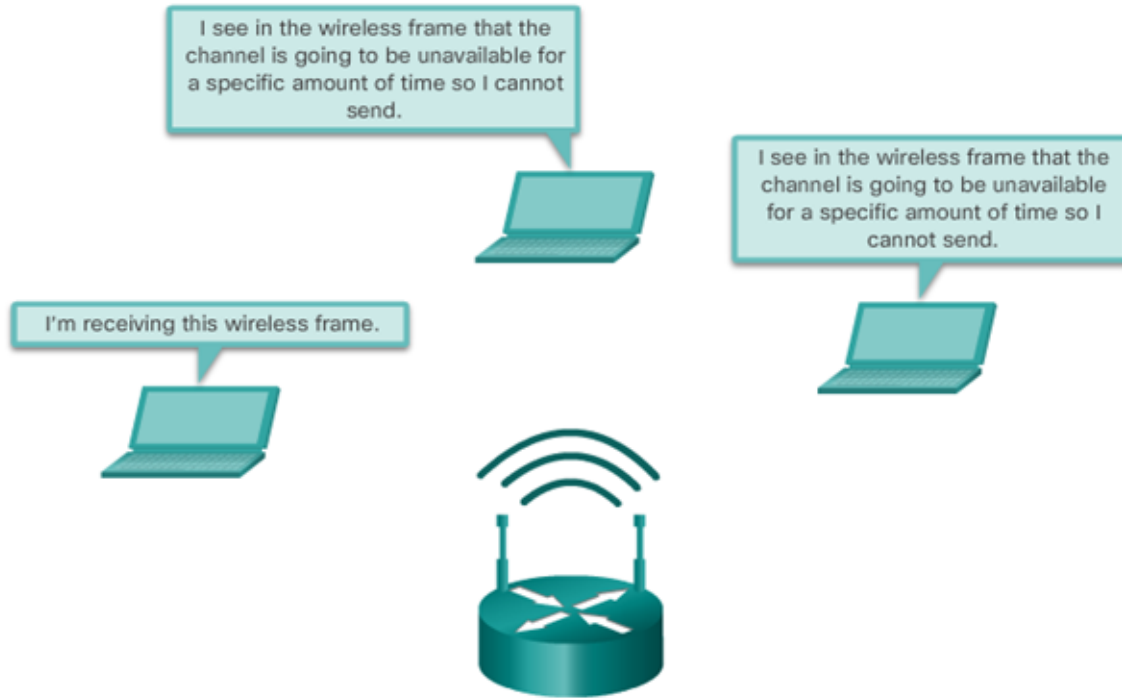
CSMA/CA

- Used by IEEE 802.11 WLANs.
- Operates in half-duplex mode where only one device sends or receives at a time.
- Uses a collision avoidance process to govern when a device can send and what happens if multiple devices send at the same time.

CSMA/CA collision avoidance process:

- When transmitting, devices also include the time duration needed for the transmission.
- Other devices on the shared medium receive the time duration information and know how long the medium will be unavailable.

Contention Based Access – CSMA/CA



6.3 Data Link Frame

Data Link Frame

The Frame

Data is encapsulated by the data link layer with a header and a trailer to form a frame.

A data link frame has three parts:

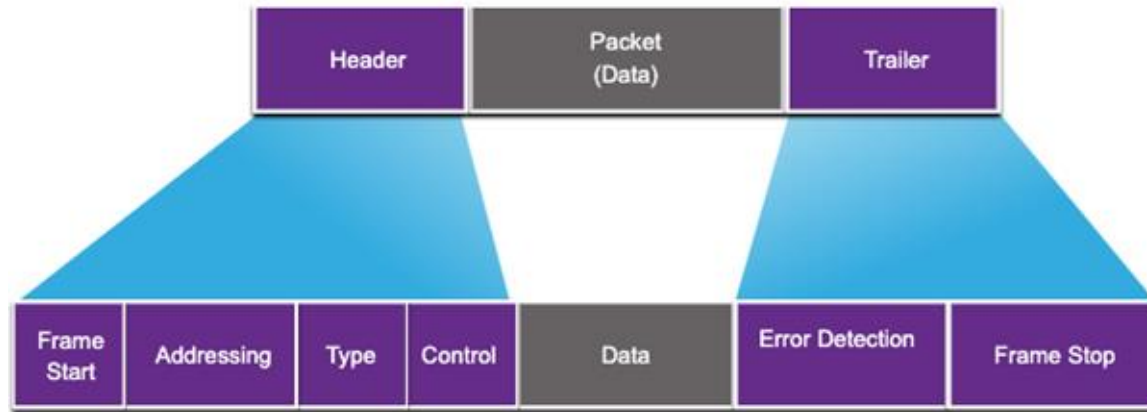
- Header
- Data
- Trailer

The fields of the header and trailer vary according to data link layer protocol.

The amount of control information carried with in the frame varies according to access control information and logical topology.

Data Link Frame

Frame Fields



Field	Description
Frame Start and Stop	Identifies beginning and end of frame
Addressing	Indicates source and destination nodes
Type	Identifies encapsulated Layer 3 protocol
Control	Identifies flow control services
Data	Contains the frame payload
Error Detection	Used for determine transmission errors

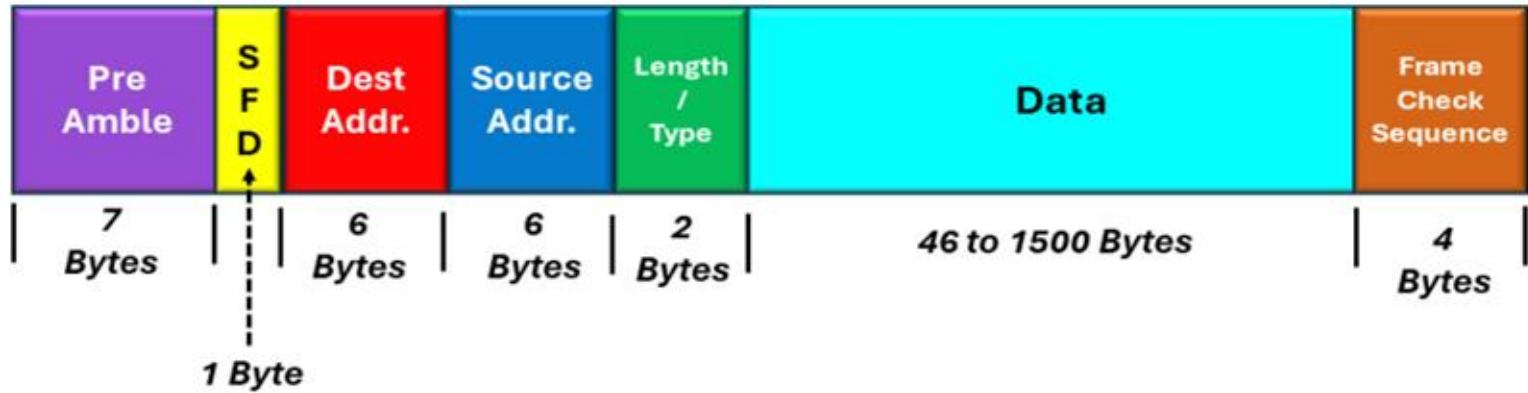
LAN and WAN Frames

The logical topology and physical media determine the data link protocol used:

- Ethernet
- 802.11 Wireless
- Point-to-Point (PPP)
- High-Level Data Link Control (HDLC)
- Frame-Relay

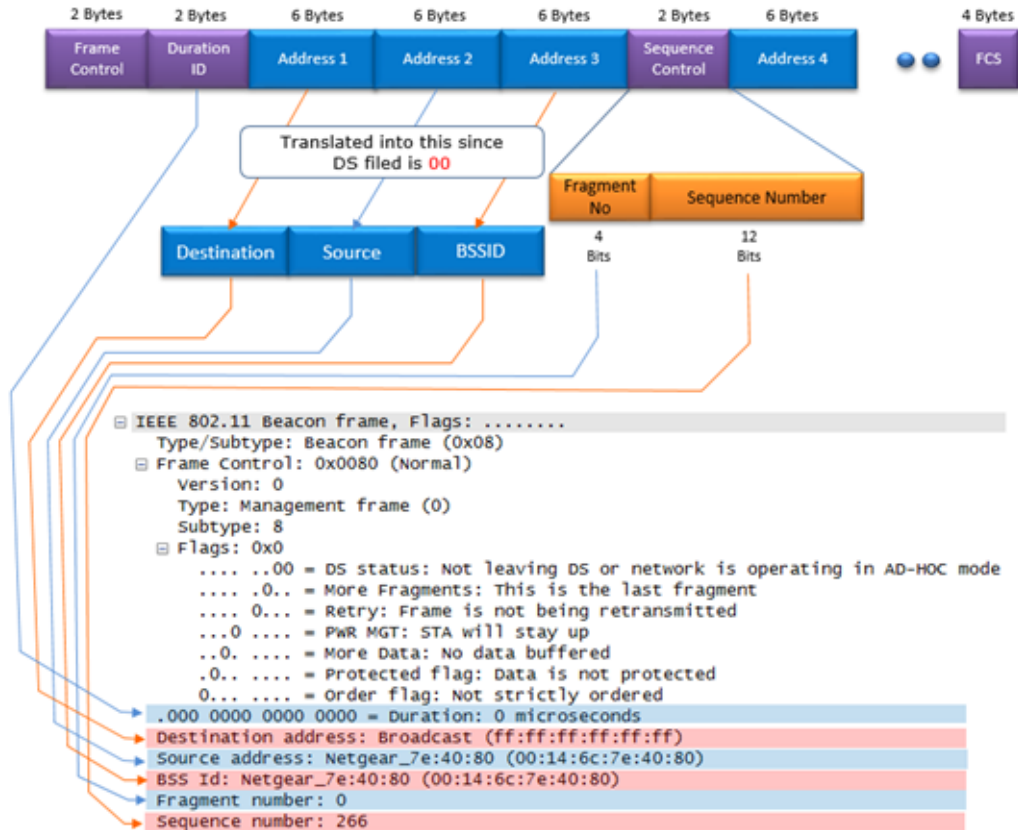
Each protocol performs media access control for specified logical topologies.

Frame Fields - Ethernet - 802.3



Minimum Frame Size = 64 Byte, Max = 1518 Bytes + Preamble

Wireless Frame - 802.11



Frame Fields - HDLC

Cisco HDLC Frame with Proprietary Type Field



Data Link Frame

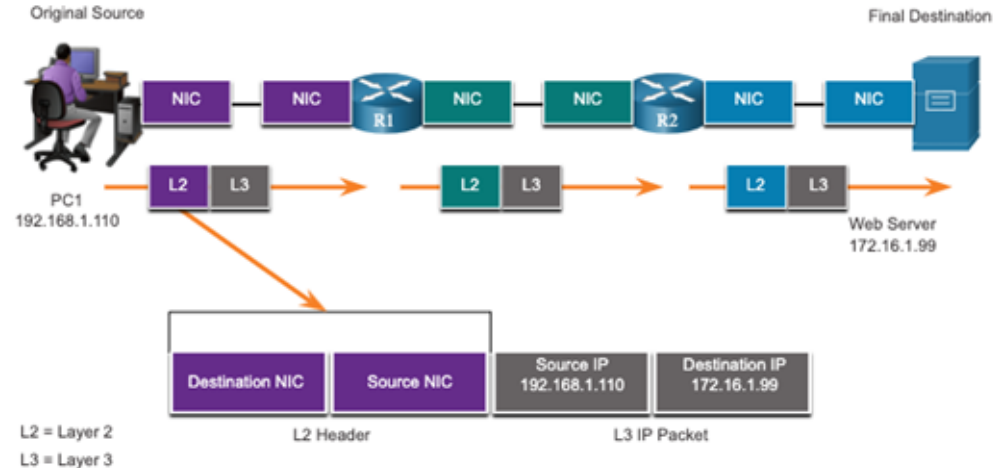
Layer 2 Addresses

Also referred to as a physical address.

Contained in the frame header.

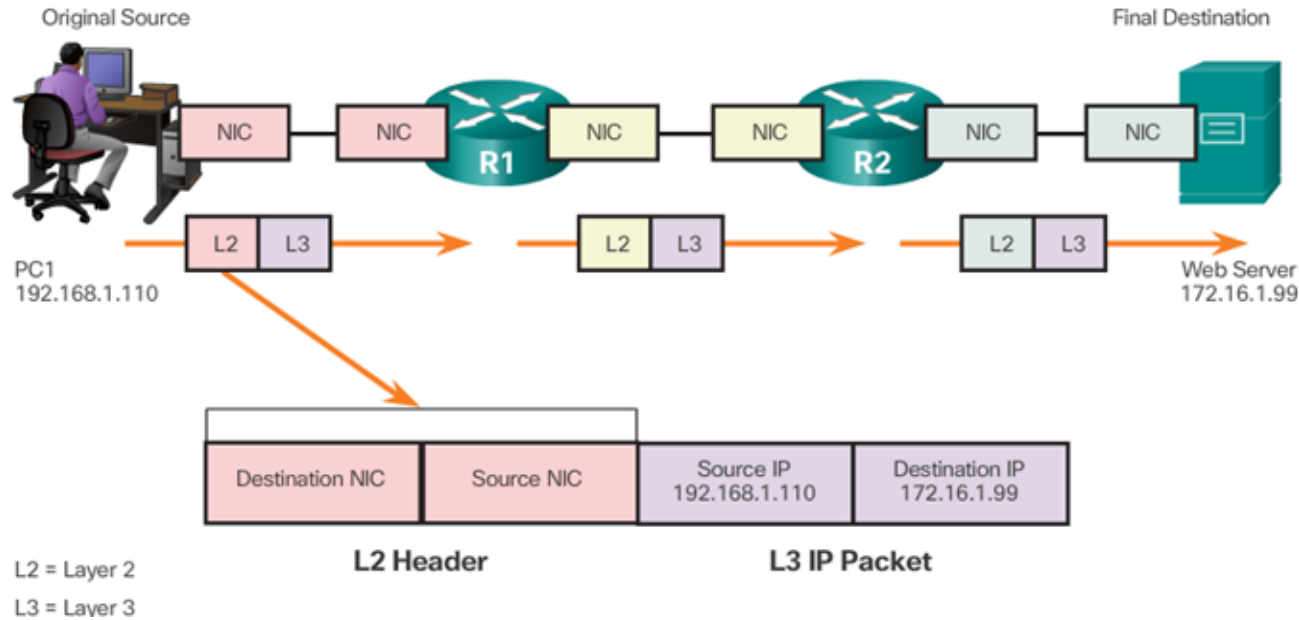
Used only for local delivery of a frame on the link.

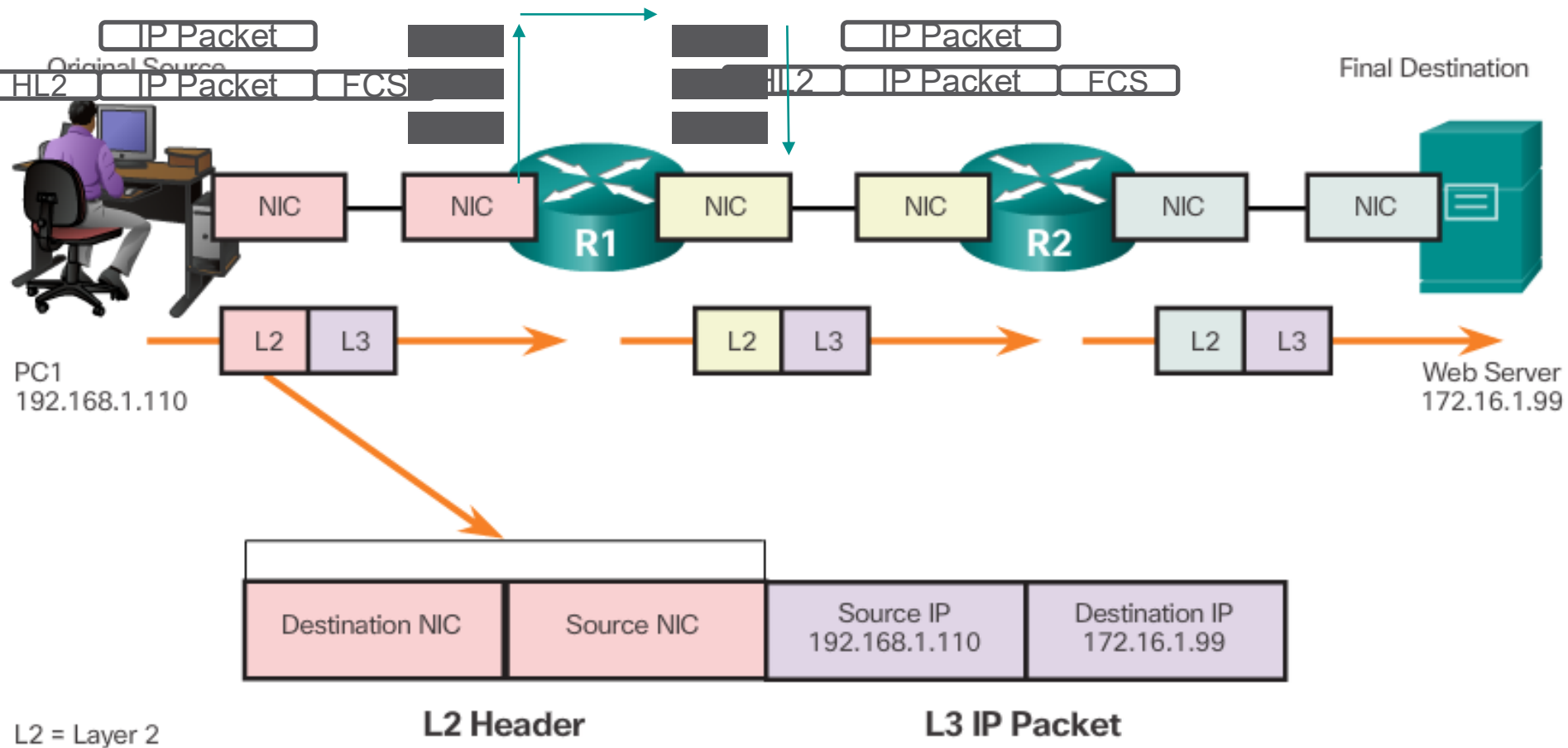
Updated by each device that forwards the frame.



Data Link Frame

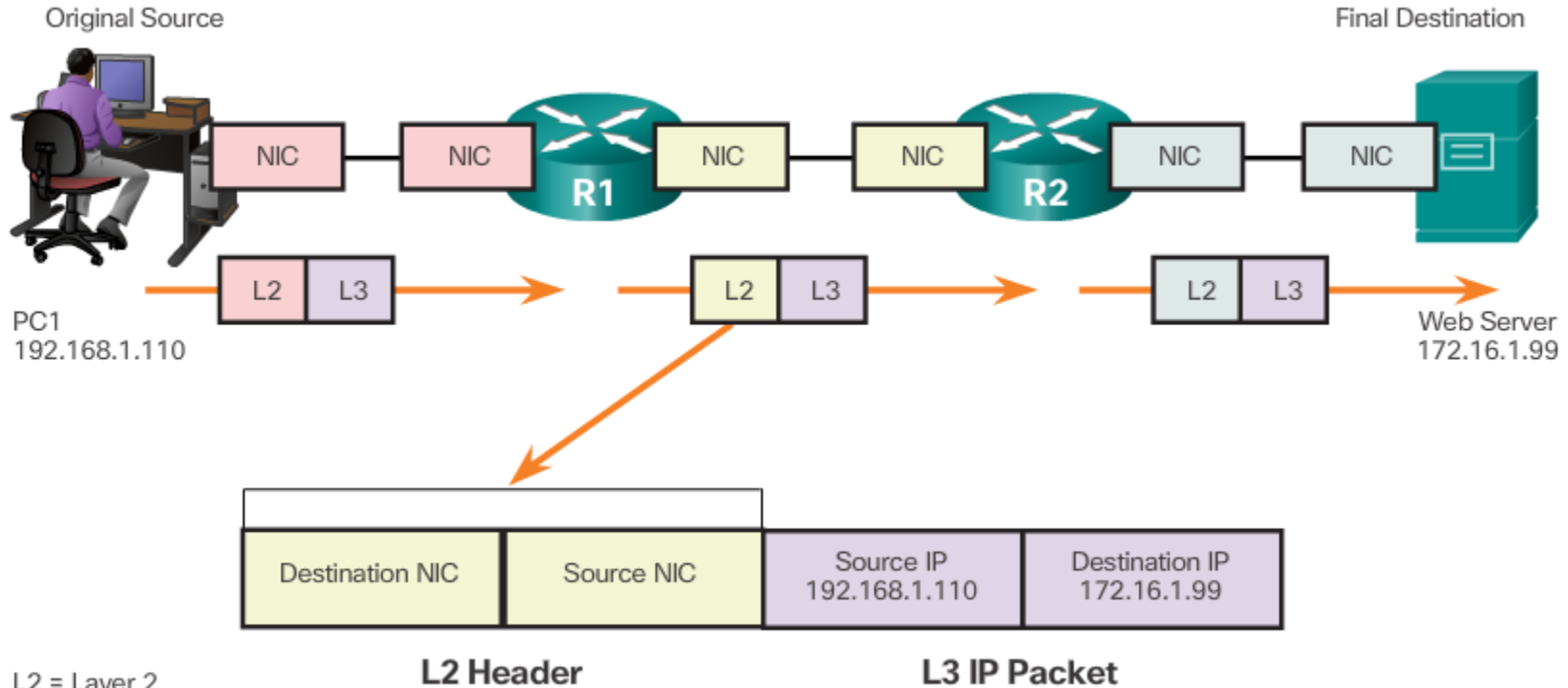
Layer 2 Addresses





Data Link Frame

Layer 2 Addresses

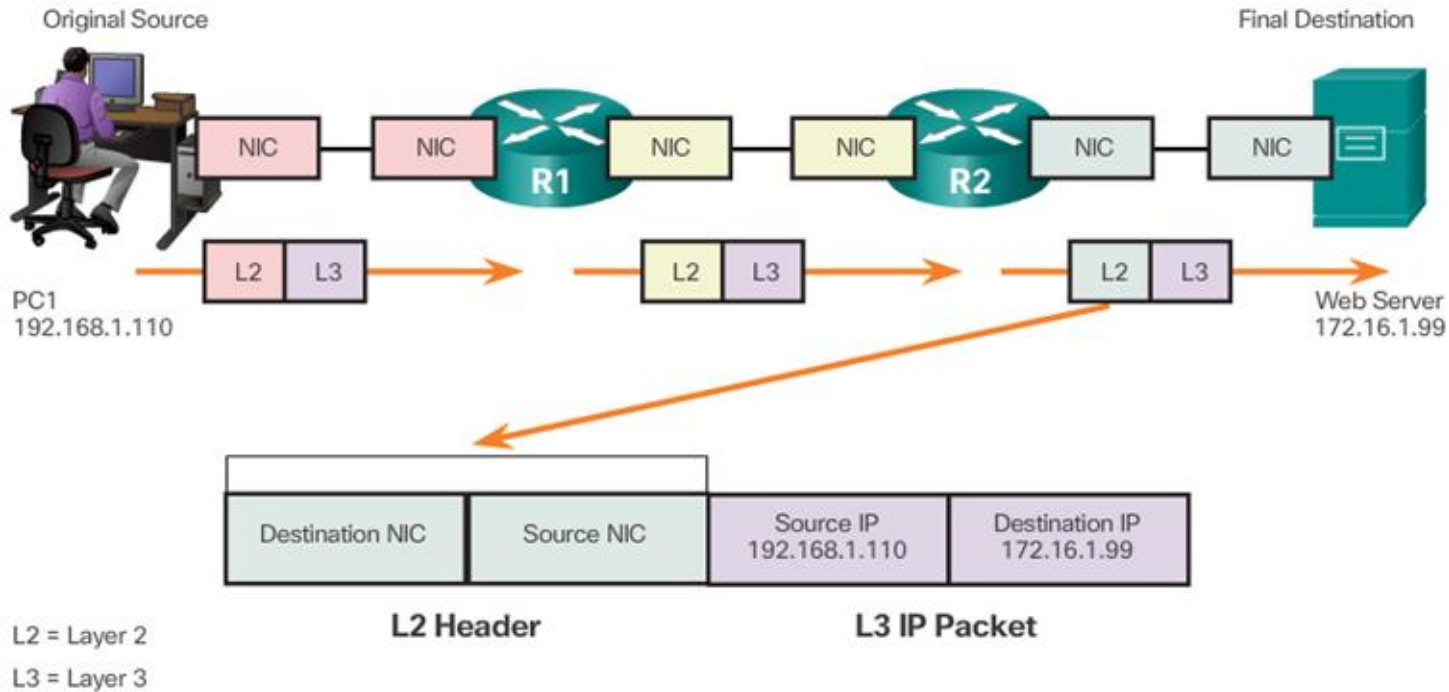


L2 = Layer 2

L3 = Layer 3

Data Link Frame

Layer 2 Addresses



9.2 ARP

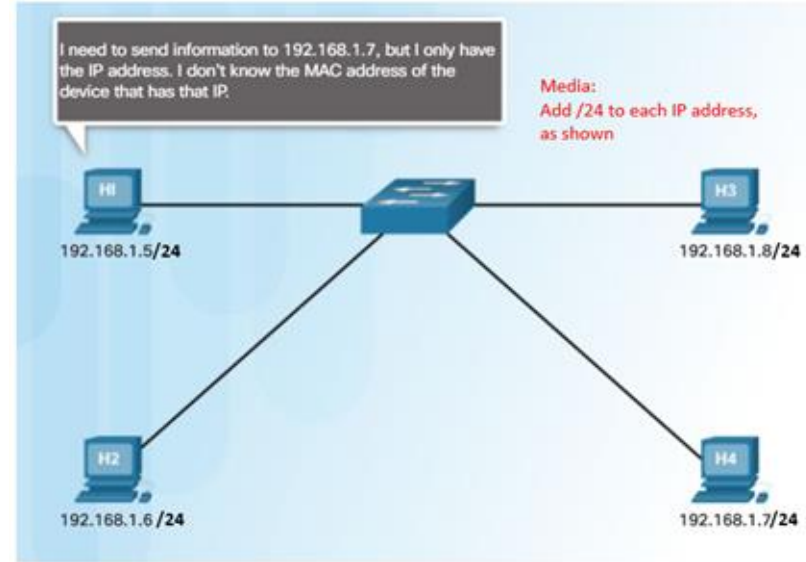
ARP

ARP Overview

A device uses ARP to determine the destination MAC address of a local device when it knows its IPv4 address.

ARP provides two basic functions:

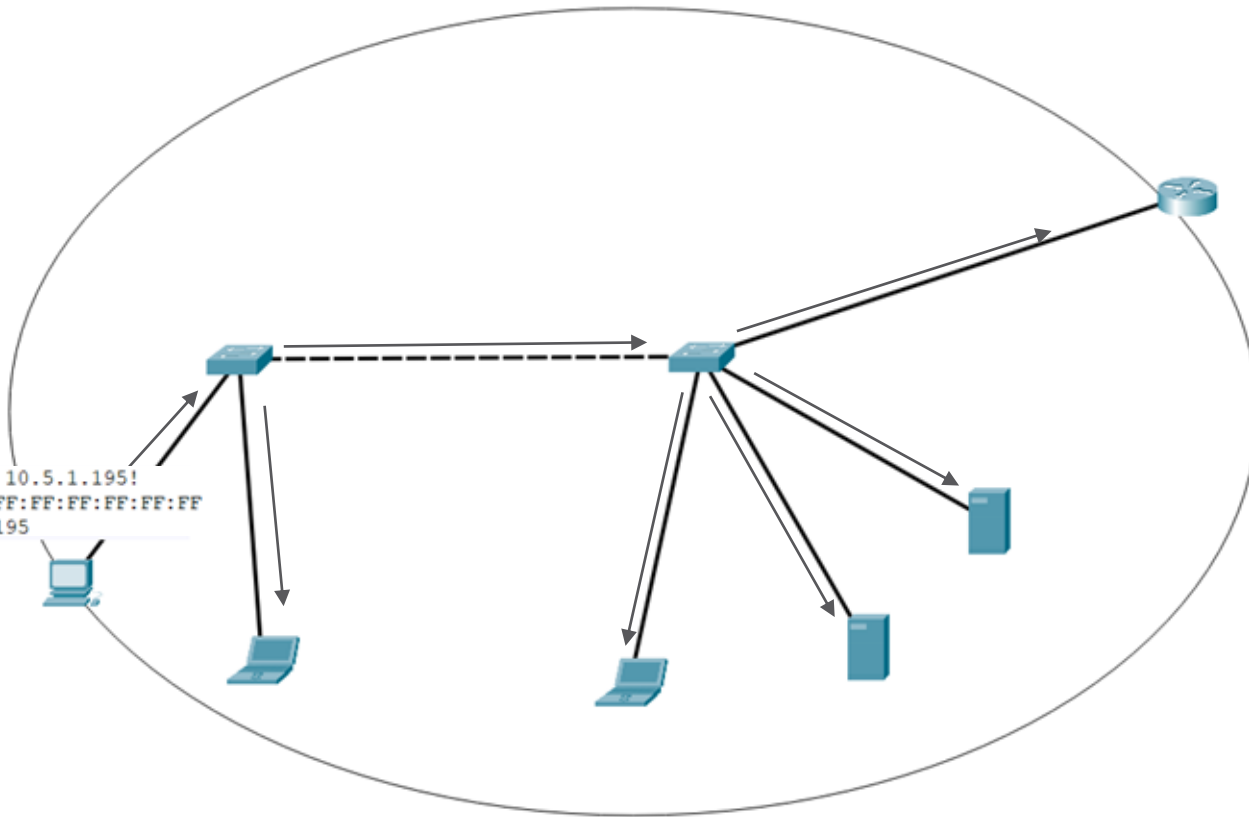
- Resolving IPv4 addresses to MAC addresses
- Maintaining an ARP table of IPv4 to MAC address mappings



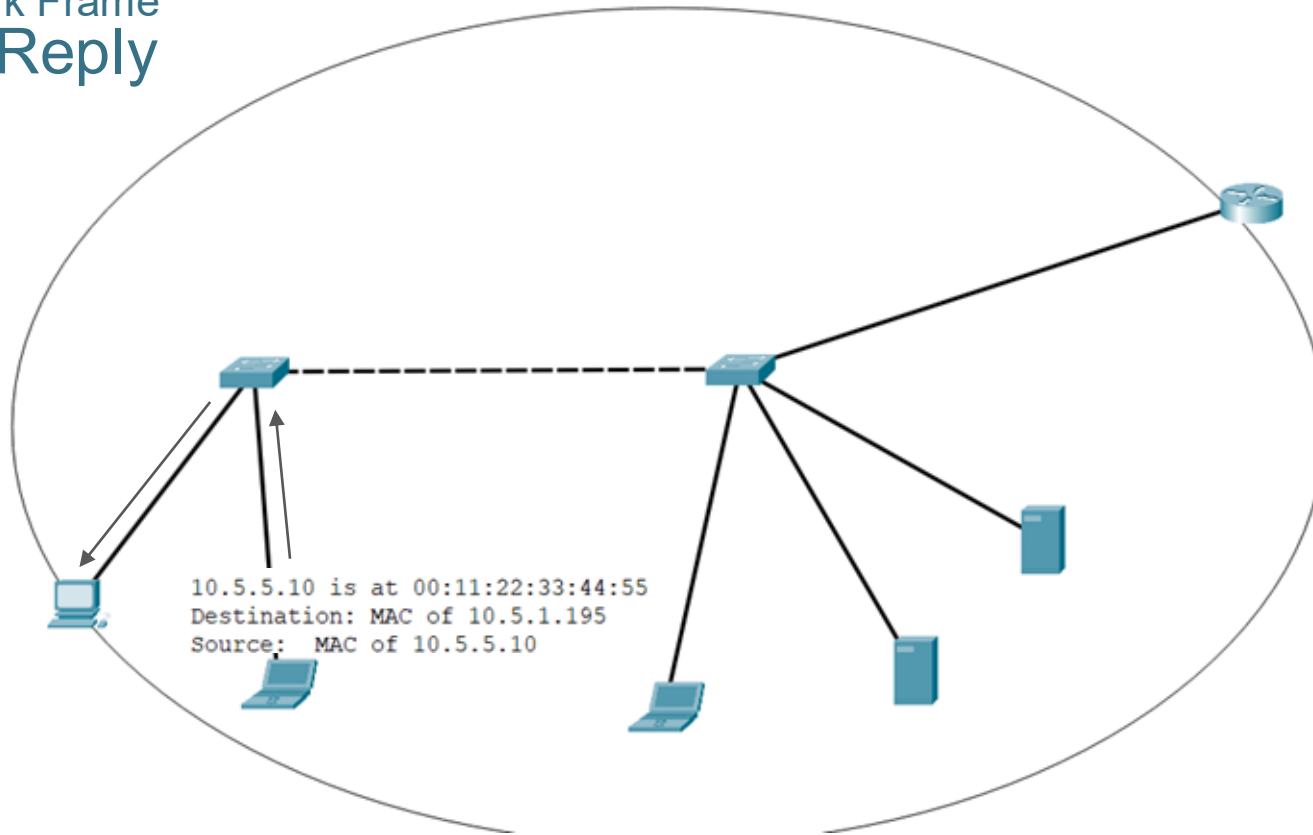
Data Link Frame

ARP

Who has 10.5.5.10? Tell 10.5.1.195!
Destination: Broadcast FF:FF:FF:FF:FF:FF
Source: MAC of 10.5.1.195



Data Link Frame ARP Reply



File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

> Frame 134: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{AAAF71DC-D9CF-4427-91B1-A5E142D68031}, id 0 > Ethernet II, Src: Intel_75:4c:0c (6c:6a:77:75:4c:0c), Dst: VMware_ef:bd:2d (00:0c:29:ef:bd:2d) > Internet Protocol Version 4, Src: 10.5.1.195, Dst: 10.5.5.10 > Internet Control Message Protocol	<pre> 0000 00 0c 29 ef bd 2d 6c 6a 77 75 4c 0c 0010 00 3c a5 b6 00 00 80 01 00 00 0a 05 0020 05 0a 08 00 3c 80 00 01 10 db 61 62 0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 0040 77 61 62 63 64 65 66 67 68 69 </pre>
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```
> Frame 133: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF_{A...}
Ethernet II, Src: VMware_ef:bd:2d (00:0c:29:ef:bd:2d), Dst: Intel_75:4c:0c (6c:6a:77:75:4c:0c)
  > Destination: Intel_75:4c:0c (6c:6a:77:75:4c:0c)
  > Source: VMware_ef:bd:2d (00:0c:29:ef:bd:2d)
    Type: ARP (0x0806)
    [Stream index: 7]
    Padding: 0000000000000000000000000000000000000000000000000000000000000000
Address Resolution Protocol (reply)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: reply (2)
  Sender MAC address: VMware_ef:bd:2d (00:0c:29:ef:bd:2d)
  Sender IP address: 10.5.5.10
  Target MAC address: Intel_75:4c:0c (6c:6a:77:75:4c:0c)
  Target IP address: 10.5.1.195
```

<<<< ARP Reply

7.2 Ethernet MAC Address

MAC Address and Hexadecimal

An Ethernet MAC address consists of a 48-bit binary value, expressed using 12 hexadecimal values.

Given that 8 bits (one byte) is a common binary grouping, binary 00000000 to 11111111 can be represented in hexadecimal as the range 00 to FF,

When using hexadecimal, leading zeroes are always displayed to complete the 8-bit representation. For example the binary value 0000 1010 is represented in hexadecimal as 0A.

Hexadecimal numbers are often represented by the value preceded by **0x** (e.g., 0x73) to distinguish between decimal and hexadecimal values in documentation.

Hexadecimal may also be represented by a subscript 16, or the hex number followed by an H (e.g., 73H).

Ethernet MAC Addresses

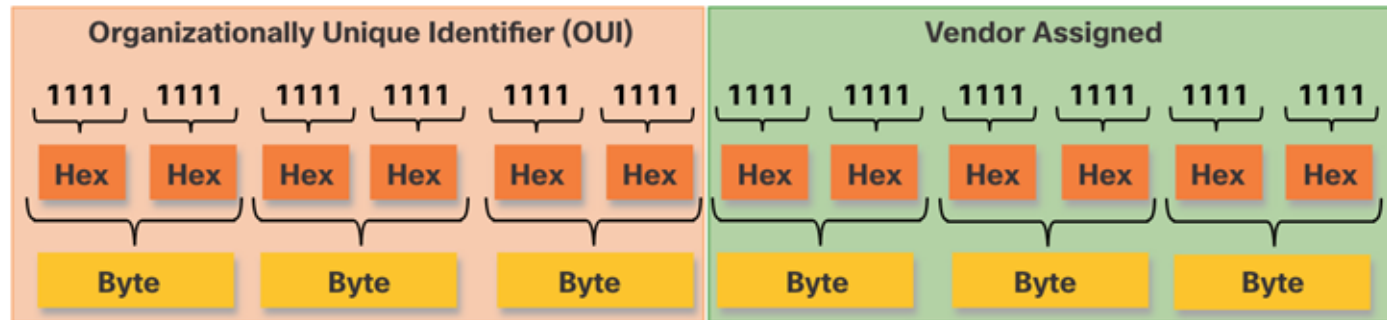
Ethernet MAC Address

In an Ethernet LAN, every network device is connected to the same, shared media. MAC addressing provides a method for device identification at the data link layer of the OSI model.

An Ethernet MAC address is a 48-bit address expressed using 12 hexadecimal digits. Because a byte equals 8 bits, we can also say that a MAC address is 6 bytes in length.

All MAC addresses must be unique to the Ethernet device or Ethernet interface. To ensure this, all vendors that sell Ethernet devices must register with the IEEE to obtain a unique 6 hexadecimal (i.e., 24-bit or 3-byte) code called the organizationally unique identifier (OUI).

An Ethernet MAC address consists of a 6 hexadecimal vendor OUI code followed by a 6 hexadecimal vendor-assigned value.



Ethernet MAC Addresses

Frame Processing

When a device is forwarding a message to an Ethernet network, the Ethernet header include a Source MAC address and a Destination MAC address.

When a NIC receives an Ethernet frame, it examines the destination MAC address to see if it matches the physical MAC address that is stored in RAM. If there is no match, the device discards the frame. If there is a match, it passes the frame up the OSI layers, where the de-encapsulation process takes place.

Note: Ethernet NICs will also accept frames if the destination MAC address is a broadcast or a multicast group of which the host is a member.

Any device that is the source or destination of an Ethernet frame, will have an Ethernet NIC and therefore, a MAC address. This includes workstations, servers, printers, mobile devices, and routers.

Destination Address	Source Address	Data
CC:CC:CC:CC:CC:CC	AA:AA:AA:AA:AA:AA	Encapsulated data
Frame Addressing		



Ethernet MAC Addresses

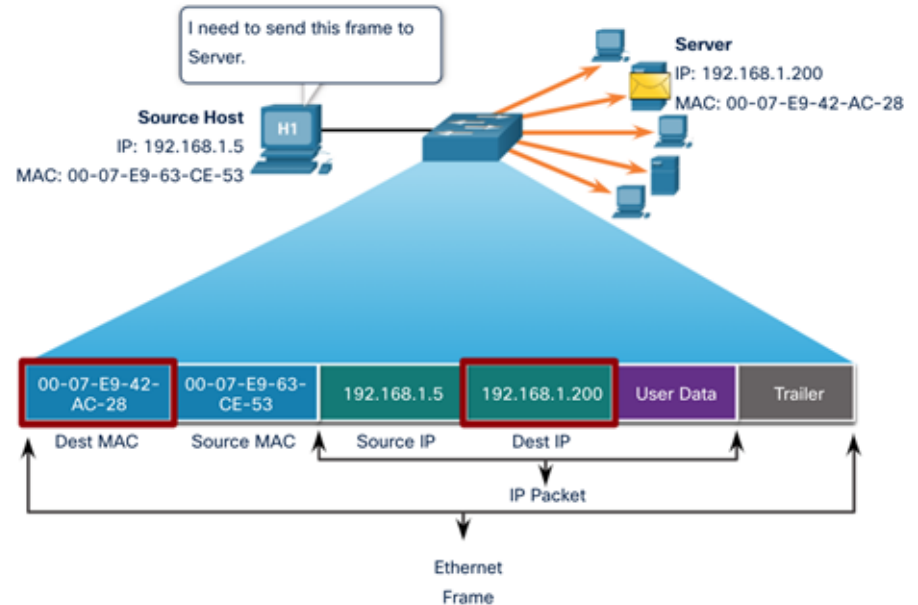
Unicast MAC Address

In Ethernet, different MAC addresses are used for Layer 2 unicast, broadcast, and multicast communications.

A unicast MAC address is the unique address that is used when a frame is sent from a single transmitting device to a single destination device.

The process that a source host uses to determine the destination MAC address associated with an IPv4 address is known as Address Resolution Protocol (ARP). The process that a source host uses to determine the destination MAC address associated with an IPv6 address is known as Neighbor Discovery (ND).

Note: The source MAC address must always be a unicast.



Ethernet MAC Addresses

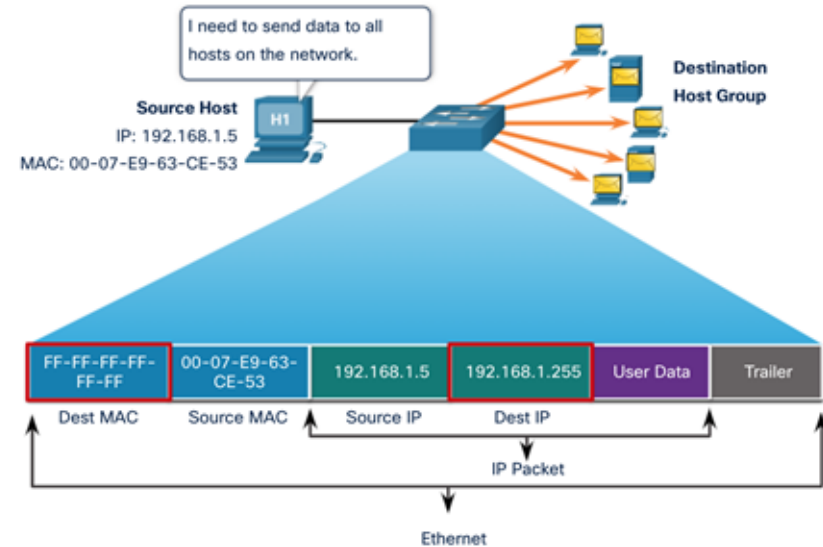
Broadcast MAC Address

An Ethernet broadcast frame is received and processed by every device on the Ethernet LAN. The features of an Ethernet broadcast are as follows:

- It has a destination MAC address of FF-FF-FF-FF-FF-FF in hexadecimal (48 ones in binary).

- It is flooded out all Ethernet switch ports except the incoming port. It is not forwarded by a router.

- If the encapsulated data is an IPv4 broadcast packet, this means the packet contains a destination IPv4 address that has all ones (1s) in the host portion. This numbering in the address means that all hosts on that local network (broadcast domain) will receive and process the packet.



Ethernet MAC Addresses

Multicast MAC Address

An Ethernet multicast frame is received and processed by a group of devices that belong to the same multicast group.

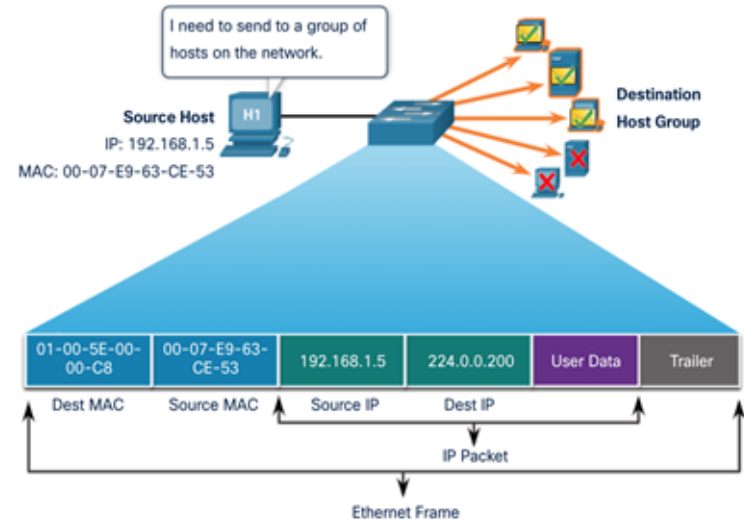
There is a destination MAC address of 01-00-5E when the encapsulated data is an IPv4 multicast packet and a destination MAC address of 33-33 when the encapsulated data is an IPv6 multicast packet.

There are other reserved multicast destination MAC addresses for when the encapsulated data is not IP, such as Spanning Tree Protocol (STP).

It is flooded out all Ethernet switch ports except the incoming port, unless the switch is configured for multicast snooping. It is not forwarded by a router, unless the router is configured to route multicast packets.

Because multicast addresses represent a group of addresses (sometimes called a host group), they can only be used as the destination of a packet. The source will always be a unicast address.

As with the unicast and broadcast addresses, the multicast IP address requires a corresponding multicast MAC address.



Lab – View Network Device MAC Addresses

In this lab, you will complete the following objectives:

Part 1: Set Up the Topology and Initialize Devices

Part 2: Configure Devices and Verify Connectivity

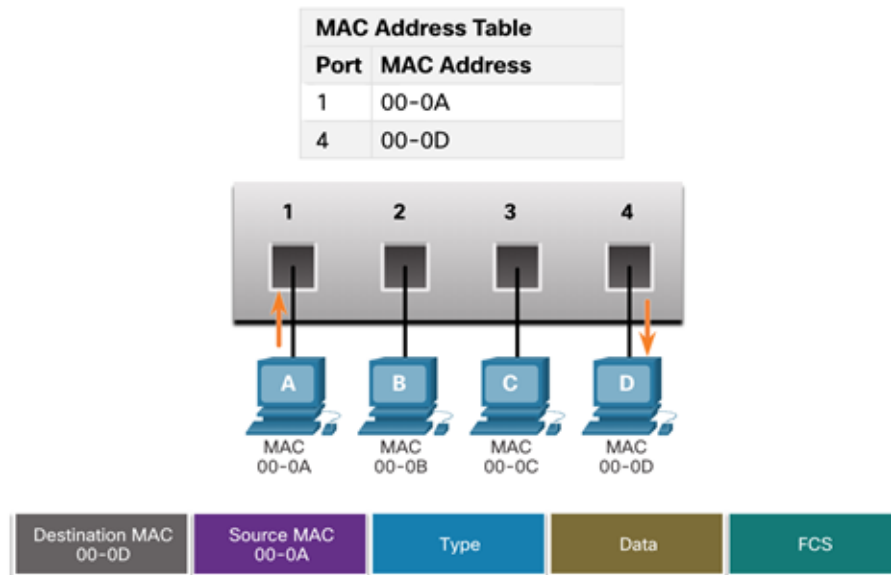
Part 3: Display, Describe, and Analyze Ethernet MAC Addresses

7.3 The MAC Address Table

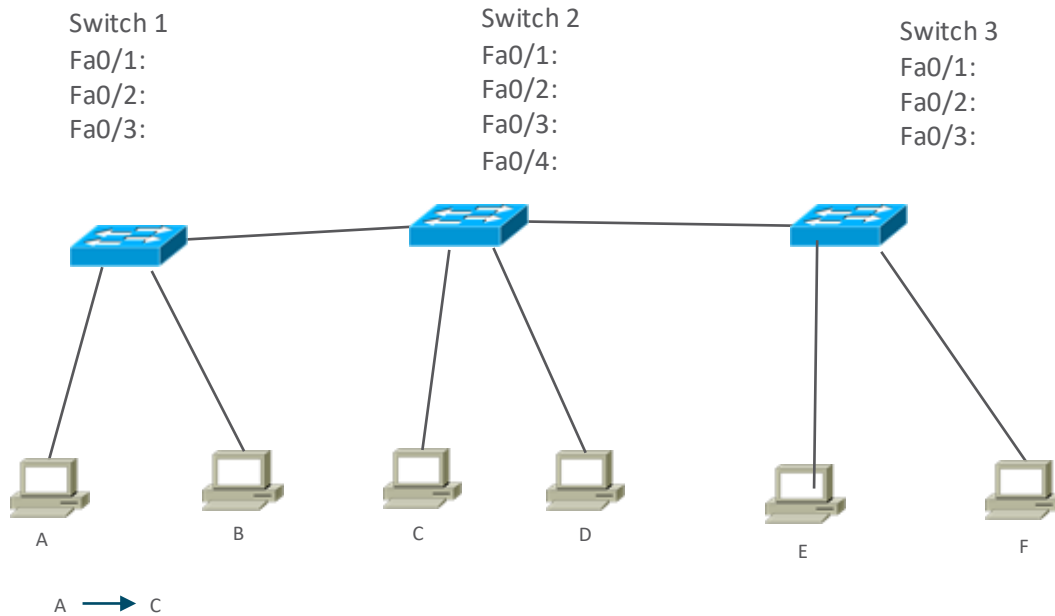
The MAC Address Table

Filtering Frames

As a switch receives frames from different devices, it is able to populate its MAC address table by examining the source MAC address of every frame. When the MAC address table of the switch contains the destination MAC address, it is able to filter the frame and forward out a single port.



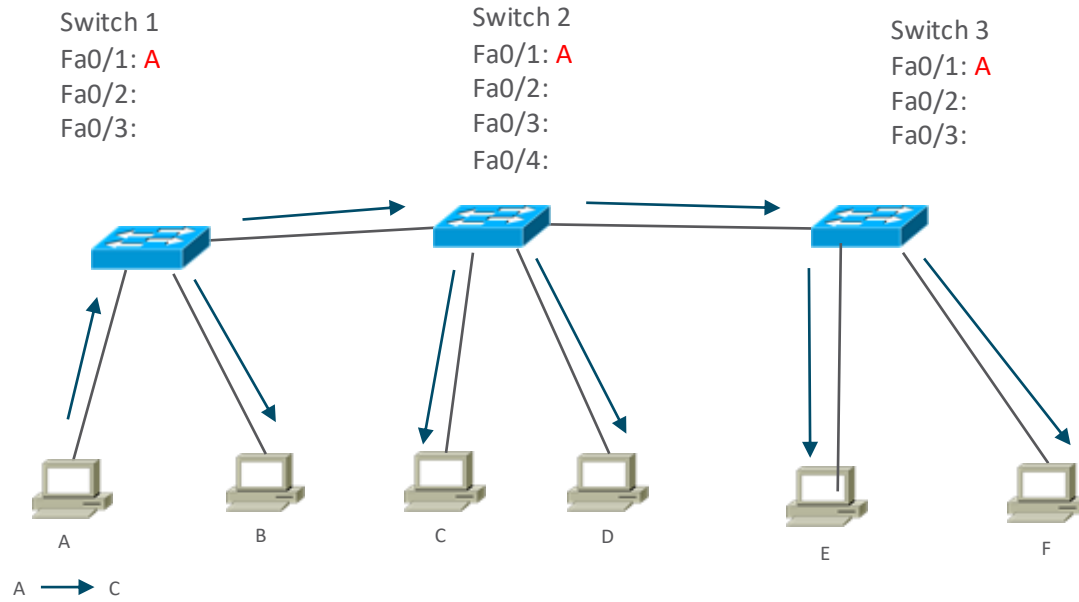
Switch Functions



Switch Functions

Flooding

Learning



Switch Functions

Flooding

Learning

Forwarding

Filtering

Switch 1

Fa0/1: A

Fa0/2:

Fa0/3: C

Switch 2

Fa0/1: A

Fa0/2: C

Fa0/3:

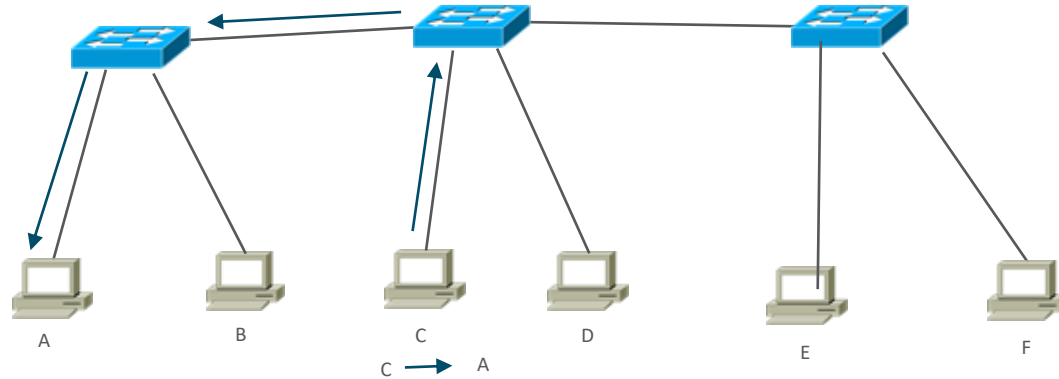
Fa0/4:

Switch 3

Fa0/1: A

Fa0/2:

Fa0/3:



Switch Functions

Flooding

Learning

Forwarding

Filtering

Switch 1

Fa0/1: A

Fa0/2: B

Fa0/3: CDEF

Switch 2

Fa0/1: AB

Fa0/2: C

Fa0/3: D

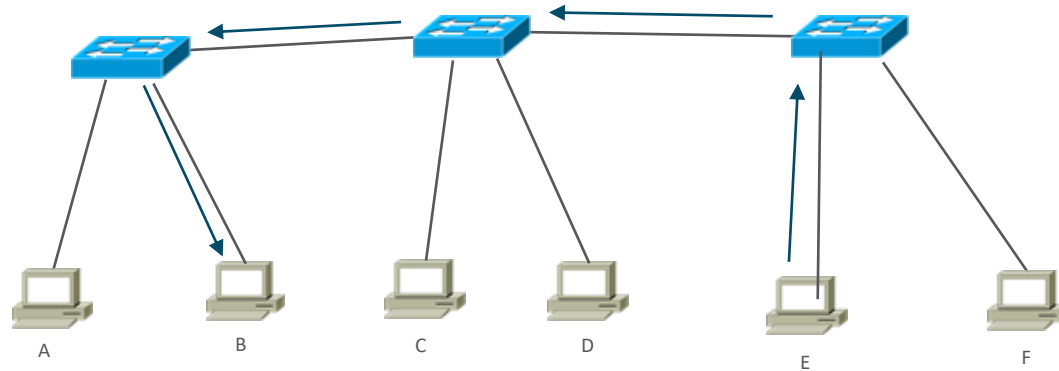
Fa0/4: EF

Switch 3

Fa0/1: ABCD

Fa0/2: E

Fa0/3: F



Switch Functions

Switch 1
Fa0/1: A
Fa0/2: B
Fa0/3: CDEF

Switch 2
Fa0/1: AB
Fa0/2: C
Fa0/3:D
Fa0/4:EF

Switch 3
Fa0/1: ABCD
Fa0/2:E
Fa0/3:F

